

InetAddress Internals

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NOT NULL
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Environment

- GNU/ Linux – Debian 13
- OpenJDK 21
- dual-stack

Memory (in)efficiency of IPv4 in Java

- For Inet4Address Java uses 56 B instead of 32 b (4 B), 15x overhead
- Inet4Address object uses 24 B currently
- InetAddressHolder 32 B and can contain hostname a address type (IPv4 = 1, IPv6 = 2)
- Straight forward to lower to ~24 B total now, with project Lilliput and Valhalla (value objects) even more, perhaps 16 B or even less

Memory (in)efficiency of IPv6 in Java

- For `Inet6Address` uses 120 B instead of 128 b (16 B), 7,5x overhead
- `Inet6Address` object uses `InetAddressHolder` inherited from `InetAddress` and adds `Inet6AddressHolder`, which has scope and an array for the address itself.
- In comparison `java.util.UUID` uses 2x long
- 32B – 40 B should be doable now

In Linux kernel

```
struct in_addr {  
    __be32 s_addr;  
};
```

```
struct in6_addr {  
    union {  
        __u8    u6_addr8[16];  
        __be16  u6_addr16[8];  
        __be32  u6_addr32[4];  
    } in6_u;  
};
```

The layout of the object

Inet@25618e91d footprint:

Table is sorted by "SUM".

Printing first 30 lines. Use -DprintFirst=# to override.

COUNT	AVG	SUM	DESCRIPTION
2	32	64	java.net.InetAddress.InetAddressHolder
1	32	32	byte[]
1	32	32	java.net.Inet6Address.Inet6AddressHolder
1	24	24	Inet
1	24	24	java.net.Inet4Address
1	24	24	java.net.Inet6Address
7	168	200	<total>

Memory usage with Inet4Address

Name	Live Bytes	Live Objects
java.net.InetAddress\$InetAddressHolder	1,600,001,120 B (52.9%)	50,000,035 (49.9%)
java.net.Inet4Address	1,200,000,432 B (39.6%)	50,000,018 (49.9%)
java.net.Inet4Address[]	200,000,016 B (6.6%)	1 (0%)

$$32 \text{ B} + 24 \text{ B} = 56 \text{ B}$$

Memory usage with Inet6Address

Name	Live Bytes	Live Objects
byte[]	1,604,062,224 B (25.8%)	50,052,790 (25%)
java.net.InetAddress\$InetAddressHolder	1,600,000,672 B (25.7%)	50,000,021 (25%)
java.net.Inet6Address\$Inet6AddressHolder	1,600,000,288 B (25.7%)	50,000,009 (25%)
java.net.Inet6Address	1,200,000,216 B (19.3%)	50,000,009 (25%)
java.net.Inet6Address[]	200,000,016 B (3.2%)	1 (0%)

$$3 \times 32 \text{ B} + 24 \text{ B} = 120 \text{ B}$$

VisualVM


```

Clojure 1.12.0
(ns user
  (:import (java.net InetAddress Inet6Address Inet4Address)))

(declare from-inet-address)

(defrecord inet4-address [^int address]

  Object
  (toString [inet4]
    (.toString (from-inet-address inet4))))

;; Should also do reader literals
(defrecord inet6-address [^long msb ^long lsb]

  Object
  (toString [inet6]
    (.toString (from-inet-address inet6))))

(defn bytes->num
  [ba index upper-bound]
  (loop [i index acc 0]
    (if (< i upper-bound)
      (recur (inc i) (+ (bit-shift-left acc 8) (bit-and (aget ba i) 0xff)))
      acc)))

(defn num->bytes
  [ba num index upper-bound]
  (loop [i index shift (* (- (dec upper-bound) index) 8)] ; 56 for long ; 24 for int presumably
    (if (< i upper-bound)
      (do (aset ba i (unchecked-byte (bit-and (bit-shift-right num shift) 0xff)))
          (recur (inc i) (- shift 8)))
      ba)))

(defn ->inet-address
  [^String inet-str]
  (let [inet (InetAddress/getByName inet-str)]
    (cond (instance? Inet6Address inet)
      (let [^bytes inet6-bytes (.getAddress ^Inet6Address inet)]
        (inet6-address. (bytes->num inet6-bytes 0 8)
          (bytes->num inet6-bytes 8 16)))
      (instance? Inet4Address inet)
      (let [^bytes inet4-bytes (.getAddress ^Inet4Address inet)]
        (inet4-address. (unchecked-int (bytes->num inet4-bytes 0 4)))))))

(defn from-inet-address
  [inet]
  (cond (instance? inet6-address inet)
    (-> ^bytes (make-array Byte/TYPE 16)
      (num->bytes (.msb inet) 0 8) ; get the first half of the address as long
      (num->bytes (.lsb inet) 8 16) ; get the second half of the address as long
      (Inet6Address/getByAddress))
    (instance? inet4-address inet)
    (-> ^bytes (make-array Byte/TYPE 4)
      (num->bytes (.address inet) 0 4)
      (Inet4Address/getByAddress))))

=> nil

```

Prototype in Clojure

Sources

<https://docs.oracle.com/en/java/javase/21/core/java-networking.html#GUID-540A1A77-3890-4303-A1EA-B51CCE58463B>

<https://github.com/netty/netty/issues/13400>

<https://docs.oracle.com/en/java/javase/21/docs/api/java.base/java/net/InetAddress.html>

<https://github.com/openjdk/jdk/blob/master/src/java.base/unix/native/libnet/Inet4AddressImpl.c>

https://github.com/openjdk/jdk/blob/master/src/java.base/unix/native/libnet/net_util_md.c

<https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/net/InetAddress.java>

<https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/net/Inet4Address.java>

<https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/net/Inet6Address.java>

<https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/util/UUID.java>

man capabilities(7)

<https://www.youtube.com/watch?v=kHJ1moNLwao>

<https://github.com/torvalds/linux/blob/master/include/uapi/linux/in.h>

<https://github.com/torvalds/linux/blob/master/include/uapi/linux/in6.h>

Thank You 🎉

For feedback and later questions:
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